

Hospital Appointment Voice Agent

An n8n backend that powers a Vapi phone voice agent for a hospital front desk — checking availability, booking, rescheduling, cancelling, and queuing callbacks, entirely by voice.

n8n

Vapi

Google Sheets

Gmail

1 Executive Summary

Hospital front desks lose time to repetitive phone calls: “Is Dr. Ahmed free Thursday?”, “I need to cancel my 3pm”, “can someone call me back”. This backend gives a Vapi voice agent four callable tools that handle exactly these requests, each backed by validation logic so the agent can never book a conflicting slot, cancel someone else's appointment, or skip caller confirmation before making a change.

2 Key Metrics

Trigger	4 webhooks called directly by Vapi during a live phone call
Voice Platform	Vapi — the caller talks to an AI agent, which calls these tools mid-conversation
Data Store	Google Sheets — an Appointments tab and a Callbacks tab
Staff Notifications	Gmail, sent on every new booking, cancellation, reschedule and callback
Confirmation Rule	Bookings and changes are rejected unless the caller explicitly confirmed them
Identity Check	Cancel/reschedule requests must match an existing appointment ID and phone number
Fallback	A callback request is logged whenever a request can't be resolved by voice

4

VOICE TOOLS

2

SHEET TABS

0

DOUBLE-BOOKINGS POSSIBLE

3 Business Value

- ✓ **Front desk staff freed up** — routine scheduling calls are handled end-to-end by the voice agent, day or night.
- ✓ **No double-bookings** — every booking is checked against the doctor's existing confirmed slots before it's accepted.
- ✓ **No unauthorized changes** — cancelling or rescheduling requires both the appointment ID and the phone number on file.
- ✓ **Staff always in the loop** — every booking, cancellation, reschedule and callback lands in their inbox immediately.

4 Architecture Overview

Vapi's voice agent is configured with four tools, each pointing at its own n8n webhook.

TOOL	CALLED WHEN	PURPOSE
check_appointment_availability	Caller asks about a slot	Checks whether a doctor already has a confirmed slot at the requested time
book_appointment	Caller confirms a booking	Saves a caller-confirmed booking to Sheets and emails staff
manage_appointment	Caller wants to cancel/reschedule	Finds an existing appointment by ID + phone, then cancels or reschedules it
request_human_callback	Request can't be resolved by voice	Logs a callback request and emails staff — not a live transfer

5 How It Works

Check Availability

- 1 Receives doctor, department, date and time from the call.
- 2 Reads every row from the Appointments sheet.
- 3 Normalizes dates/times regardless of format ("3pm" or "15:00").
- 4 Replies with a clear yes/no the agent can speak directly to the caller.

Book Appointment

- 1 Validates that all required fields are present and the caller explicitly confirmed.
- 2 Generates a unique appointment ID and standardizes the record.
- 3 Appends the appointment to Sheets as confirmed and emails staff the full details.
- 4 Confirms back to the caller with the doctor, date, time and appointment ID.

Manage Appointment

- 1 Validates the action (cancel/reschedule), appointment ID, phone number and confirmation.
- 2 Searches for an active appointment matching both the ID and phone number.
- 3 If nothing matches, tells the caller instead of guessing which booking they meant.
- 4 Cancels or merges the changed fields, updates the sheet, and emails staff.
- 5 Confirms back with a status of cancelled, rescheduled, or not_found.

Request Callback

- 1 Captures patient name, phone number, reason and the call ID.
- 2 Appends the request to a separate Callbacks tab as pending.
- 3 Emails staff the patient's details so a human can call back.
- 4 Confirms to the caller that the request was sent — never transfers the live call.

6 Reliability & Security Design

- ◆ **Explicit confirmation required** — booking or changing an appointment is rejected unless the caller's confirmation flag was set.
- ◆ **Two-factor lookup for changes** — cancelling or rescheduling requires both the exact appointment ID and the phone number on file.
- ◆ **Server-side conflict checking** — availability is verified against the sheet at booking time, not left to the AI to infer.
- ◆ **Input validation on every tool** — each webhook checks its required fields before doing anything.
- ◆ **Full audit trail** — every booking, cancellation, reschedule and callback is logged to Sheets and emailed to staff.
- ◆ **Safe fallback** — when a request can't be resolved by voice, the agent logs a callback instead of leaving the caller stuck.

7 Setup Checklist

- ☒ Connect a Google Sheets credential with an Appointments tab and a Callbacks tab.
- ☒ Connect a Gmail credential and set the staff notification address on the four "Send Staff" nodes.
- ☒ Activate the workflow and copy each webhook's Production URL into the matching Vapi tool.
- ☒ Configure Vapi to pass caller_confirmed only once the caller has verbally agreed.
- ☒ Test all four tools end-to-end with a live or simulated call.

8 Node-by-Node Map

check_availability	Appointment Availability Webhook → Read Appointments → Check Doctor Slot → Respond to Vapi
book_appointment	Book Appointment Webhook → Validate Booking → Normalize Data → Log to Sheet → Send Staff Email → Respond to Vapi
manage_appointment	Manage Appointment Webhook → Validate Request → Find Matching Appointment → Cancel/Reschedule → Respond to Vapi
request_human_callback	Callback Request Webhook → Normalize Data → Log to Sheet → Send Staff Email → Respond to Vapi

SUMMARY

Designed and built end-to-end in n8n, demonstrating a voice-agent backend with real-time conflict checking, identity-verified appointment changes, and staff notifications.